



MENDIX TRAINEE – SERVICE POS PROJECT

AN INTERNSHIP REPORT

DURATION

12.07.2022-18.08.2022

Titan Company Limited

Integrity – 193 Veerasandra

Electronic City, Bangalore-560100.

Submitted by
Varsha Sundararaj

ABSTRACT

Mendix is a new technology that lets you to create multi-channel apps that deliver great user experiences on any device. User interface the design is designed with creative integrated Mendix framework that is fully responsive and extensible. It allows you to use app in store by online and by cash in World of Titan(WOT) showroom. They proposed “Titan Care POS app service for Multi-user” it is methodology, tool or app where the customers can enter the details and it get fetch from the database and proceed with any special occasions and process the quick billing method, if the product is in damage means it can enter complaint and the watch engineers can repair it and proceed with invoice generation and pay with cash or online. It ensures quality of products, spares and monitoring of customers database with complaints to watch engineers. The main contribution of this report is to develop the process of quick billing, Multi-billing, Service job order, goods receipts, etc., in a single platform from ideation phase to application phase. It may help the customer to monitor the progress of service and even can say their choice in online and avoid being in punctual. Engineers can be sending the message customers that their product is ready through color code. It provides tools to build, test, deploy, and iterate applications. It is based on domain model associations and reference selector on visual and model-driven development. It offers both ordering (business domain experts) and repairing (professional technical developers) the products.

1. Introduction

1.1 Objective of the Project

The objective of this project is to build a mobile and web app for the naive applications using the framework used in the World of Titan Showroom (WOT) to order the watches in-store online by the customer, and they can pay by online method or cash.

1.2 Scope

This application has following features:

- User can see all the data related to watches.
- Users can login and choose their own product of choice.
- User can submit their customer details, get invoice and pay online or cash.

2. SYSTEM ANALYSIS

2.1 Overview of Existing System

Microsoft Dynamics is a line of Business Applications, containing enterprise resource planning (ERP) and customer relationship management (CRM). Microsoft markets Dynamics applications through a network of reselling partners who provide specialized services. Microsoft Dynamics forms part of "Microsoft Business Solutions". Dynamics can be used with other Microsoft programs and services, such as **SharePoint, Yammer, Office 365, Azure and Outlook**. The Microsoft Dynamics focus-industries are retail, services, manufacturing, financial services, and the public sector. Microsoft Dynamics offers services for small, medium, and large businesses.

On-premises vs. Cloud-based POS Systems

- On-premises POS systems ideally operate within the internal network of an organization, and use local servers in storing information. POS tools are

purchased up front, with data accessible only from the local server or through an alternate web app.

- Cloud-based POS systems, on the other hand, leverage cloud technology to provide users with the ability to access the platform from anywhere and at any time. They work well with most POS hardware, and offer a level of comfort that is unparalleled.

On-premises POS system

Advantages	Disadvantages
Extensive display setup	Annual license
Diverse functionalities	Expensive to implement and maintain
Easy-to-use	Poor mobile optimization
High security	Not suitable for large businesses

Cloud-based POS system

Advantages	Disadvantages
Automated data entries	Internet-dependent
High accuracy, scalability, security.	Prone to service disruptions
Inventory management	Expensive to customize
Subscription-based pricing	Limited customer support
Remote Access	Not suitable for small businesses.

Configuring of Microsoft Dynamics POS

Microsoft Dynamics 365 has an integrated POS system that allows organizations to unify their digital, in-store, and back-office operations and create exceptional, insightful shopping experiences.

It offers two types of POS experiences:

- a browser-based cloud POS (cPOS) - that can be used on mobile devices, and
- a retail modern POS (mPOS) - that can be used on PCs, tablets, and phones to process sales transactions, customer orders, and daily operations, as well as to perform inventory management.

Uses of Microsoft Dynamics POS

- **Define Payment Methods:** helps you set up the payment methods that will be supported by your retail businesses. Since customers are likely to shop from any retail channel including brick and mortar, e-commerce and call centres, your POS system should be configured such that it supports all channel types.
- **Define Card Types:** types of cards customers might use at a POS system including credit, debit, loyalty, or gift cards. To ensure seamless payment through cards in your retail business, you must first determine the types of cards you plan to accept.
- **Setup Registers:** define your register include hardware profile (physical configuration of a terminal) and visual profile (screen characteristics), store number, and sales menu.
- **Setup New Stores:** to activate a new store, all you need to do is select the store from the drop-down list, enable full data sync, and select the distribution schedule you want to run.
- **Define Devices:** type of application you want to use, validate the devices for activation, and follow the instructions to activate your device.

2.2. Advantages and Disadvantages

Advantages

- Customization based on User/Business needs to cater business/work process.
- The adaptation of business/work processes based on user/business demands.

- Integration with other Microsoft Products.
- Capability for automated process with AI support.

Disadvantages

- The UI is not very user-friendly while being generally user-friendly as a CRM application.
- Though it has, overall user-friendly as a CRM Application interface is not very user friendly.
- Data storage capacity is limited.



2.3 Proposed system

Mendix is the app platform for building Enterprise applications. With Mendix you can Develop web based as well as mobile based application without even writing a single of code. Mendix Platform is designed to accelerate enterprise app delivery across your entire application development lifecycle, from ideation to deployment and operations. With Mendix you can even forecast your business. Mendix provides PAAS (Product as a Service) on which you can develop custom application as per your requirement.

Mendix offers both no-code and low-code tooling in one single fully integrated platform:

1. No code – Mendix provides a web-based visual app-modelling studio that is tailored toward business domain experts
2. Low code – an extensive and powerful desktop-based visual app-modelling studio is also provided, which is tailored towards professional developers and can be integrated with coding IDEs to extend capabilities

Features of Mendix

- Web App & Mobile App Development.
- Cloud infrastructure for App hosting.
- Multi-experience.
- Artificial Intelligence, Intelligent Automation.
- Data Integration, Workflow Automation, IOT based
- Extend your Application with Custom Java, JavaScript, HTML code.
- Deployment Option for Public as well as
- Private Cloud, Fast Deployment
- Flexible with Databases
- Developers can easily manage repositories.

Stages in Mendix – Why is useful?

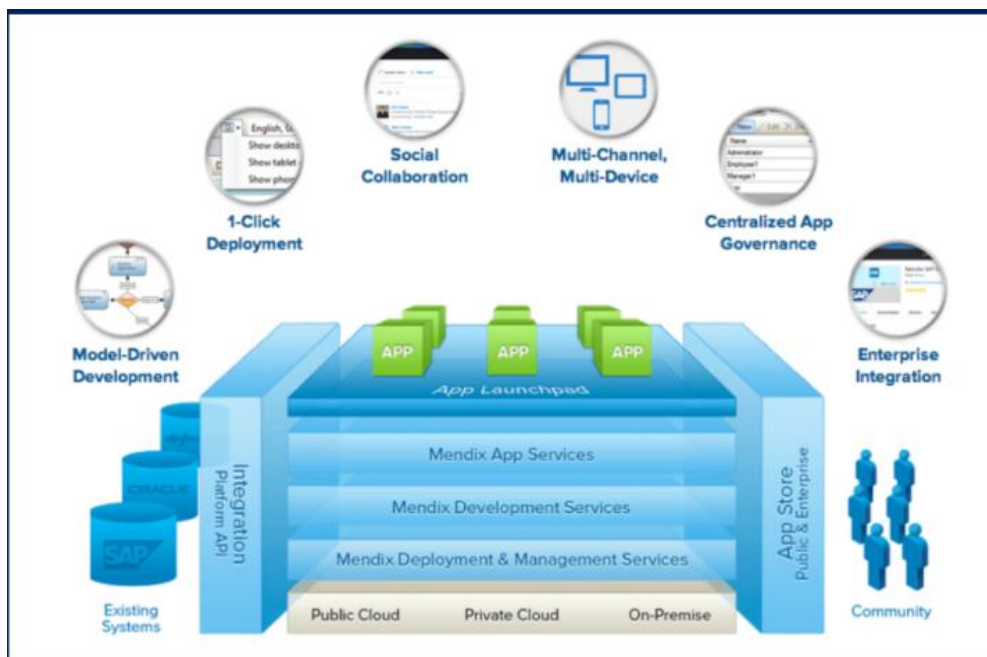
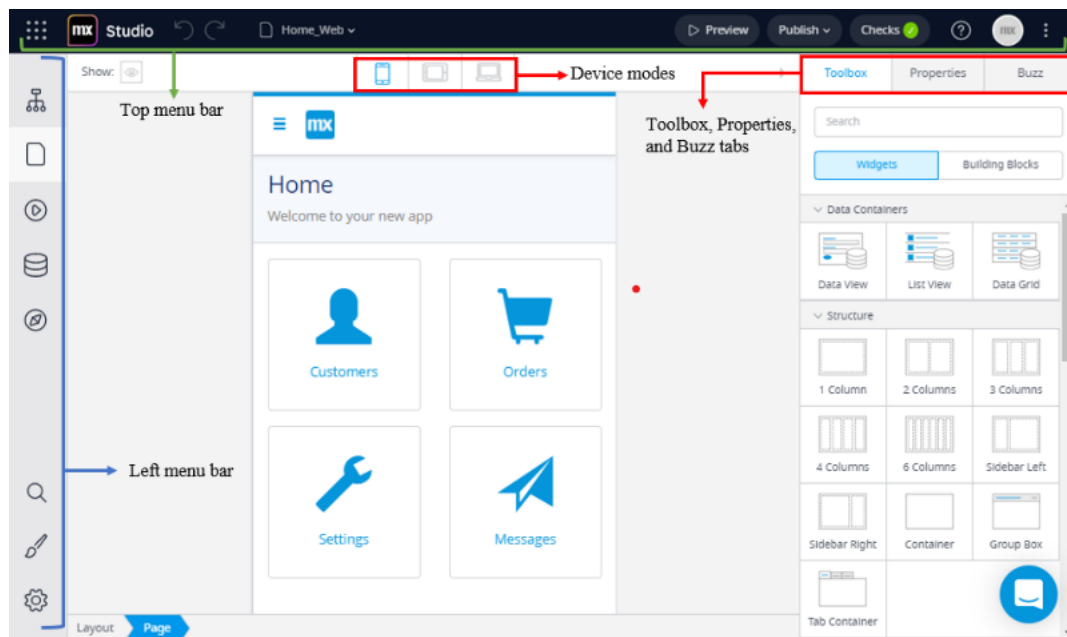


Diagram Contains

- Device Models – Mobile, Tablet & Responsive (Desktop)
- Top menu bar
- Toolbox, properties and buzz tabs
- Left menu bar – Pages, Domain model, microflows, workflows, Navigation Document, Theme Customizer & settings etc.,



Mendix includes: Client Activities & Object Activities

Client Activities – microflow and nanoflow

- Nanoflow call – calls another nanoflow
- Close page – closes the currently open page
- Download file – downloads a file using the browser
- Show home page – opens the home page for the end-user
- Show message – shows a message to the end-user
- Show page – shows a selected page to the end-user
- Synchronize to device – enables selectively synchronizing a set of objects to the device when calling a microflow from a nanoflow
- Synchronize – synchronizes data between your device and the server
- Validation feedback – does a validation check, and if this check fails, it shows a message to the end-user

Object Activities – microflow & nanoflow

- Cast object – changes the type of object from a generalized object type to the specialized object type
- Change object – changes the members of an object
- Commit objects – either stores objects of persistable entities in the database or stores objects of non-persistable entities in memory to allow them to be rolled back
- Create object – creates an object
- Delete object – activity deletes one or more objects
- Retrieve – gets one or more objects of an entity
- Rollback object – undoes uncommitted changes made to an object.

Microflow

Microflows allow you to express the logic of your application. A microflow can perform actions such as creating and updating objects, showing pages and making choices.

- Events - represent start and endpoints of a microflow and special operations in a loop.
- Flows - form the connection between elements.
- Decisions - deal with making choices and merging different paths again.
- Activities - are the actions that are executed in a microflow.
- Loop - is used to iterate over a list of objects.
- Parameter - is data that serves as input for the microflow.
- Annotation - is an element that can be used to put comments in a microflow.

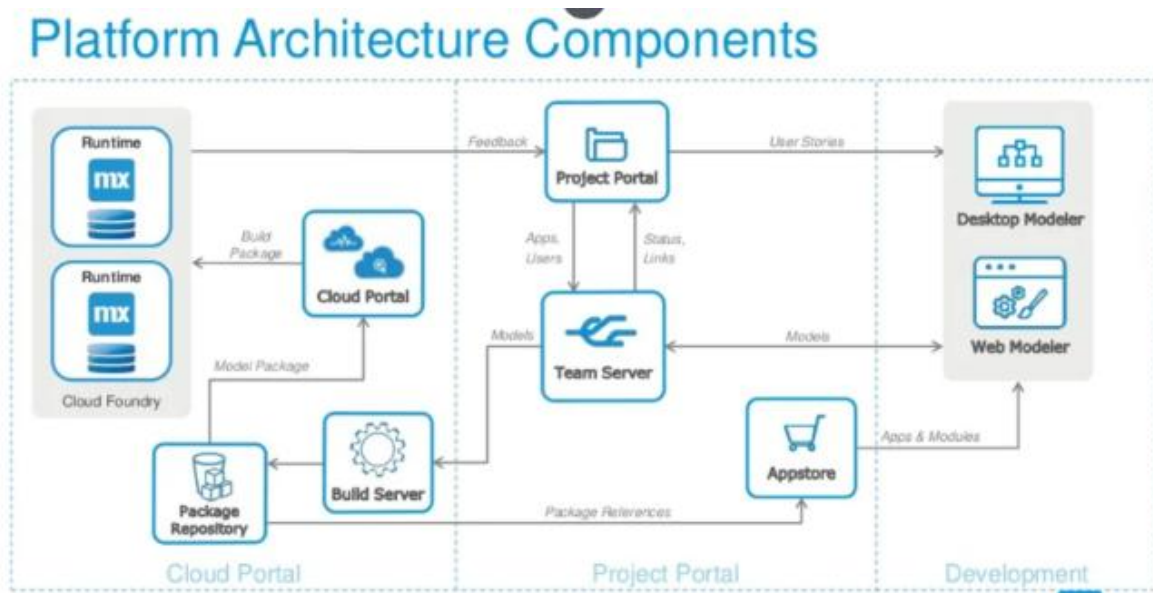
Nanoflows

Nanoflows are similar to microflows, in that they allow you to express the logic of your application.

- Events - represent start and endpoints of a microflow and special operations in a loop.
- Flows - form the connection between elements.
- Decisions - deal with making choices and merging different paths again.

- Activities - are the actions that are executed in a microflow.
- Loop - is used to iterate over a list of objects.
- Parameter - is data that serves as input for the microflow.
- Annotation - is an element that can be used to put comments in a microflow.

2.4 Architecture of Mendix



3. SYSTEM DESIGN

3.1 Introduction

The Mendix tool was used for the course due to its functionality, simplicity and the support offered by Mendix. The platform is the leading low-code solution recognized by analyst reports like Gardner and large enterprise companies such as SAP and IBM. It is a full stack platform designed to build applications rapidly.

The platform abstracts and automates the various application development layers from front end to back end. The logic uses diagrams called unified modeler Language(UML), business logic uses Business Process Modeling Notation (BPMN) and the user interface is built with widgets following a what you see is what you get (WYSIWYG) model.

The platform allows for business and IT to collaborate and build applications that add business value.

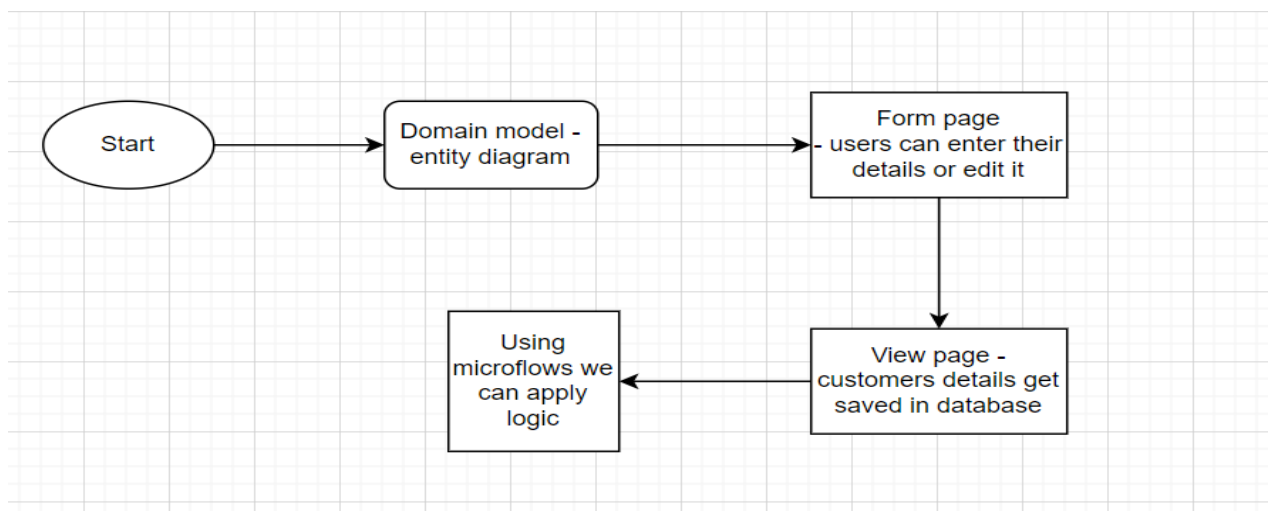
It handles 6 main functions:

- Collaboration
- Data Structure and Domain Model
- Business Logic
- User Interface and Experience
- Security and user authentication
- Deployment

Project Ideas

- An app to keep track shopping cart purchased by customer.
- Tracking board games and ratings.
- A Simple ticketing system that allows to raise tickets and manage them.
- We can get the student mark with total in microflow.
- Online stationary shop.

3.2 Architecture – Mendix



3.3 Concepts

3.3.1 Quick Billing

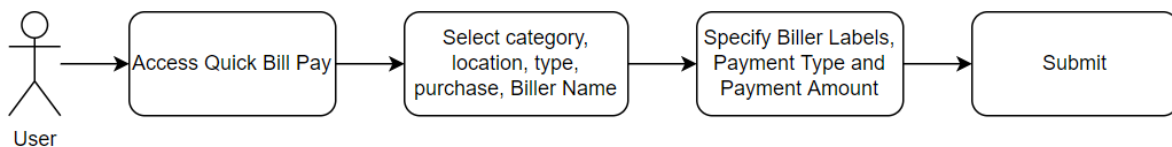
- The Quick Bill payment feature enables the users to pay their utility bills online without registration. Through this feature, users are able to pay their bills quickly, securely and at their own convenience.
- The user can initiate bill payment transactions to 'Payment' and 'Presentment and Payment' type of billers who allow Quick payment.

- The user also has an option to enter the payment amount, select the recharge plan, the source of funds from which payment can be made, and the date on which payment can be made.

Precondition:

- Retail user is provided with Transaction access
- Transaction working window and billers are maintained.
- Limits are assigned to the user to perform the transaction process.

Workflows – Quick billing



3.3.2 Multi billing

Multiple Billing means a temporary indirect cost rate applicable to a specified period that is used for funding, interim reimbursement, and reporting indirect costs on federal or federal pass-through awards pending the establishment of a final rate for the period.

Features of Multi billing

- Billing & invoicing
- Customer Accounts
- Carrier Access Billing
- Self Service Portal
- Service Order System

3.3.3 Goods Receipts

- A goods receipt, depending on the context, has two different meaning.
 - It is the movement of goods and materials into the warehouse
Eg. Incoming goods
 - It is a document that serves as a confirmation to the receipt of materials to the warehouse or the receipt of stock from the vendor or manufacturer.
- It is a function of purchasing goods in material Management or the entry to goods receipt accounts in Financial Accounting.

- Companies to ensure the replenishment of semi-finished products, raw materials, supplies and finished products use it.

Features of Goods Receipts

- Goods receipt referencing a purchase order
- Goods receipt not referencing a purchase order
- Goods receipt referencing a delivery
- Rough goods receipts, Support for store goods receipts.
- Entry of returnable transport packaging as special stock (third-party stock)
- Reversal of goods receipts and posting of returns to supplier
- Entry of quantities for variants of generic articles using a matrix.

4. Work responsibilities

4.1 Internship position

As an intern in Titan Company Limited, I worked in IT team where the team checks and provides overall services to watches in stores (WOT). I worked as a developer Trainee where I should build app for customers to order watches in online, which includes:

Customer details: Customer information that have collected.

Quick billing: records were created automatically when a user adds a customer to quick billing, and an appropriate quick billing record does not already exist.

Multi billing: the method used when Switched Exchange Services is provided by the Parties

Goods receipt: information from a purchase order or stock transport order (STO) in SAP, and the physical materials and services are matched to the vendor invoice and purchase order.

4.2 Description of Internship

The internship started with a 1-month training period which gave the new experience in working with the new platform Mendix – no code, low code which is new launch to the IT industry. After the training program, I was provided with the project to build the app with my own skills. I had a regular guidance with **Ms.**

Paarkavi who has an experience in this Mendix platform with the idea of business logic.

First, we need to construct a domain model which an entity diagram where we associate with other entity, data stored in the database and information of customer details with quick billing with job order no, complaint and spares are added to it and at last the invoice generation is processed.

The Mendix have in-built code like Java, CSS, HTML, Mysql, etc. The respective microflow can be created with the tools, buttons by using the expressions like query and even validation can be added, and the result can be attained easily.

Microflow



4.3 Project – screenshots

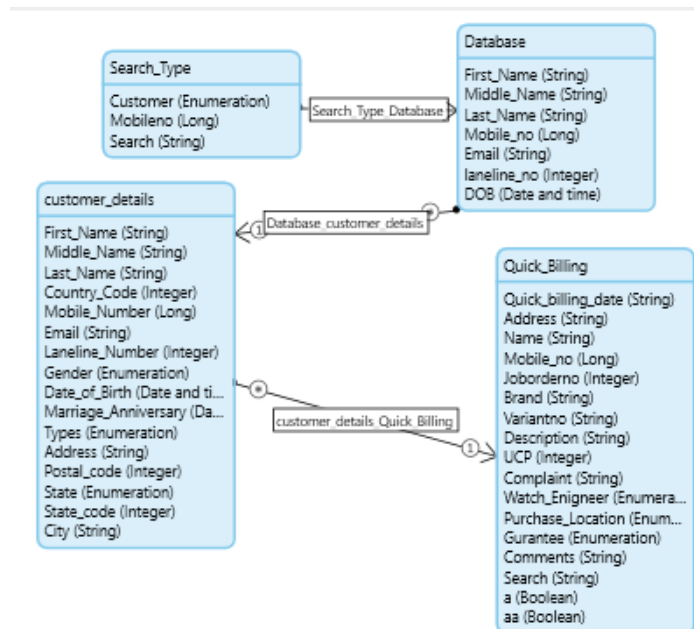
Home Page

The screenshot displays the Mendix IDE interface. On the left, the 'App Explorer' shows the project structure with 'Home_Web' selected. The main canvas shows the 'Home' page design with a 'Welcome to your new app' message. On the right, the 'Properties' panel shows the configuration for the 'Home_Web' page, including 'Name', 'Documentation', 'Class', 'Style', 'Dynamic classes', 'Designer' (Canvas width: 1200, Canvas height: 600), 'General' (Platform: Web, Layout type: Responsive, Layout: Atlas_Default (Atlas_Core), Title: Homepage), 'Navigation' (Visible for: User), and 'Usage' (Mark as used: No). The bottom console shows the log of the application, including messages about updating the application model and adding request handlers.

Date/time	Log node	Message
2022-08-18 17:49:55.219	Core	Updating application model...
2022-08-18 17:49:55.263	Core	Application model has been updated, application is now available.
2022-08-18 17:49:55.267	Connector	Added request handler 'debugger/' with class 'com.mendix.modules.debugger.inter...
2022-08-18 17:51:47.418	Core	Updating application model...
2022-08-18 17:51:47.458	Core	Application model has been updated, application is now available.
2022-08-18 17:51:47.461	Connector	Added request handler 'debugger/' with class 'com.mendix.modules.debugger.inter...

Domain Model

- Domain Model
- From page
- View page
- microflow



- Search
- Database
- Customer details
- Quick billing
- Add variant no, complaints, etc,
- Add spares
- Invoice generation
- Cash method

Search

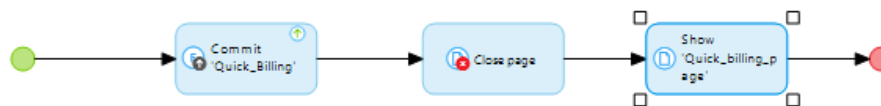
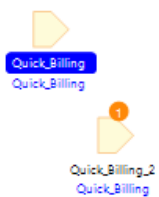
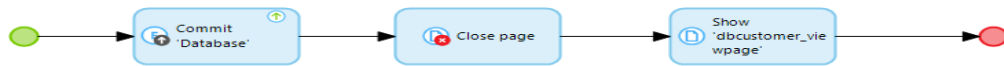
First, we need to search phone number in quick search then it gets stored in database. If he/she is present in the database means the app will fetch the data's and then it gets autofill in name, phone number, mail id, date of birth, etc.

The screenshot shows the 'QUICK SEARCH' interface. It includes a sidebar with a home icon and a main search area. The search area has a 'Search_Type' dropdown menu, a 'Customer' dropdown menu, a 'Mobileno' input field with the value '0', and a 'Search' button.

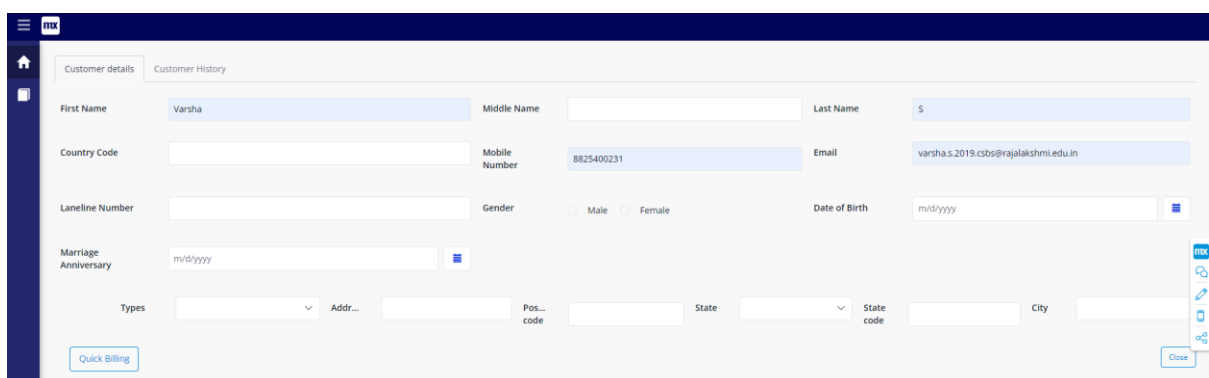
Get stored in database or it just fetch the data.

First Name	Middle Name	Last Name	Mobile no	Email	laneline no	DOB
varsha		S	8825400231	vara234@gmail.com	443442	8/26/2022
ritika		S	8834567890	ritika1232GMAIL.COM	43423	1/4/2005

Microflows



Customer details page



Customer details Customer History

First Name	Varsha	Middle Name		Last Name	S
Country Code		Mobile Number	8825400231	Email	varsha.s.2019.csbs@rajalakshmi.edu.in
Laneline Number		Gender	☐ Male ☐ Female	Date of Birth	m/d/yyyy
Marriage Anniversary	m/d/yyyy				
Types		Addr...		Pos... code	
		State		State code	
		City			

Quick Billing Close

Quick billing Page

Job Order

Association

Quick billing date

Address

Name

Mobile no

Total Charges

Spare Charges

0.00

Labour Charges

0.00

Total

0.00

Joborderno

Brand

Variantno

Description

UCP

Complaint

Watch Enigneer

Watch_Engineer

Purchase Loc...

Purchase_Location

Gurantee

Comments

Save

Cancel

Complaints

[Quick_Billing, page parameter]

Search [Search]

20%	20%	20%	40%
Selected	ProductType	Complaint code	Description
☐ [a]	Watches	7	Day indicator panit peel off
☐ [aa]	Watches	12	battery

Save

Invoice

[Company Name]

[Street Address]

[City,State,Pincode]

phone :

BILL TO

[Name]

[Company Name]

[street Address]

[City,State,Pincode]

[Phone]

INVOICE#

2034

CUSTOMER ID

504

DATE

5/8/2022

TERMS

Net 30 Days

SHIP TO

[Name]

[Company Name]

[street Address]

[City,State,Pincode]

[Phone]

S.NO	DESCRIPTION	QTY	AMOUNT
a)	TH79676110j Battery - 367	1	385
b)	Battery - 375	1	119
Thank you for your business		Total	504

If you have any questions about this invoice, please contact
[Name, Phone, email@address.com]

3.3 Outcomes of Internship

The following are the outcomes of internship:

1. Understand the basics in domain model – to create the entity model.
2. To speed up own innovations using the aesthetic colour of choice
3. Learn the importance of microflows and built our own logic.
4. Get to know more about the trending applications and easy to learn or use compared to android studio or visual studio.
5. Understand the way to proceed the business logic in naïve apps and in expressions and microflows.

4. SKILLS LEARNED

- Ability to master various web technologies and techniques
- Creating applications for enterprises or businesses using the Mendix platform
- Familiarity with Agile and DevOps

Agile - Agile methodology is a way to manage a project by breaking it up into several phases.

DevOps - the combination of cultural practices and tools that increases an organization's ability to deliver applications and services at high velocity

- Provide solutions that can help improve business value using microflows
- Database and design – to build the data models and stored it in the database like view page.
- Skilled in adopting new technology to meet application requirements.
- To provide day-to-day support for applications.
- How to use microflows in business with own logic.
- More about widgets, atlas core UI, Xi path constraints to retrieve data and many more techniques.

5. CONCLUSION

I did my internship as a “Project Trainee - Mendix” in Titan Company Limited. Mendix is a low code development platform where it brings the outstanding innovation for the development of smart applications. I have gained more knowledge in this new platform Mendix about how to use the decisions or microflows in the business point of view or in business logic. I also came to know about new widgets like client activities, object activities, nanoflows, etc., and how to design or built the queries in different languages and create ui’s based on our choice and how it is used in frameworks of sap and in business. I also learned the importance of this trending platform and how it will be used in future.

Also this project has enabled me to get exposure to new programming languages like java, JavaScript, expressions, etc, and new development platform like Microsoft dynamics & Mendix.

This internship has been an excellent and rewarding experience. I can conclude that there have been a lot I have learnt from my work at the company. Needless to say, the technical aspects of the work I have done are not flawless and could be improved on experience.

=====THANK YOU=====